

THE TREE OF ADDITION: GROWING WITH NUMBERS

1

THE ROOTS: What is Addition? (Combining Parts)



Part: 3



Part: 2



Whole: 5

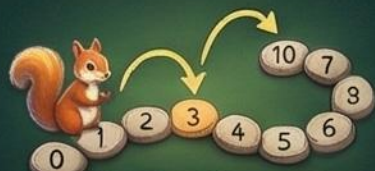
$$\rightarrow 3 + 2 = 5$$

Combining two or more numbers to find a total.

2

THE TRUNK: Visual Models

(a) Number Line (Jumps Forward)



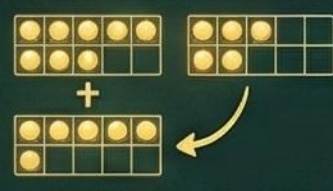
$4 + 3 = 7$ jumps
Moving forward on a path.

(b) Counters (Combining Groups)



$5 + 4 = 9$
Putting groups together.

(c) Ten-Frames (Filling the Grid)



$7 + 5 = 10 + 2 = 12$
Making a ten to add.

3

THE BRANCHES: Key Properties (Rules of Growth)

(a) Commutative Property (Order Doesn't Matter)

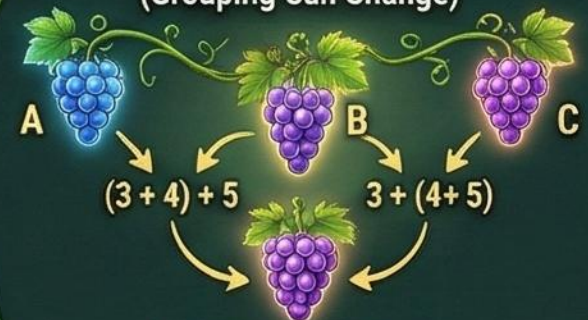


$$2 + 6 = 8$$

$$a + b = b + a \text{ (e.g., } 2 + 6 = 6 + 2)$$

$$6 + 2 = 8$$

(b) Associative Property (Grouping Can Change)



$$(a + b) + c = a + (b + c)$$

(e.g., $(3+4)+5 = 3+(4+5)$)

4

THE FRUITS: Real-World Harvest (Applications)

Counting Money



$$\text{Total} = 3 + 4 = 7 \text{ coins}$$

Adding Time



$$\text{Total Time} = 2 \text{ hrs} + 3 \text{ hrs} = 5 \text{ hrs}$$

Measuring Length



$$\text{Total Length} = 5 \text{ m} + 7 \text{ m} = 12 \text{ m}$$

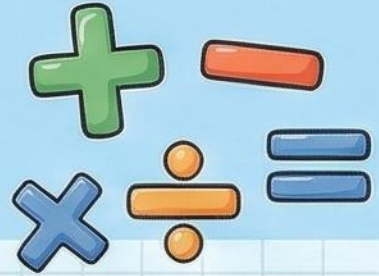
5

THE TOOL: The Grower's Chart (Addition Table)

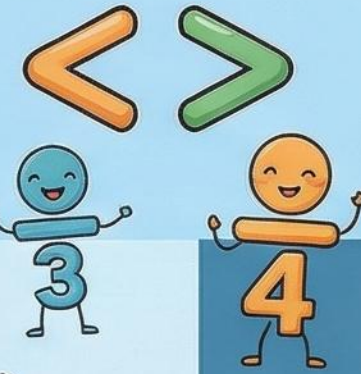
1	2	3	4
3	6		
4	5	6	14
7	8		
7	8	9	10

A helpful reference for quick sums.

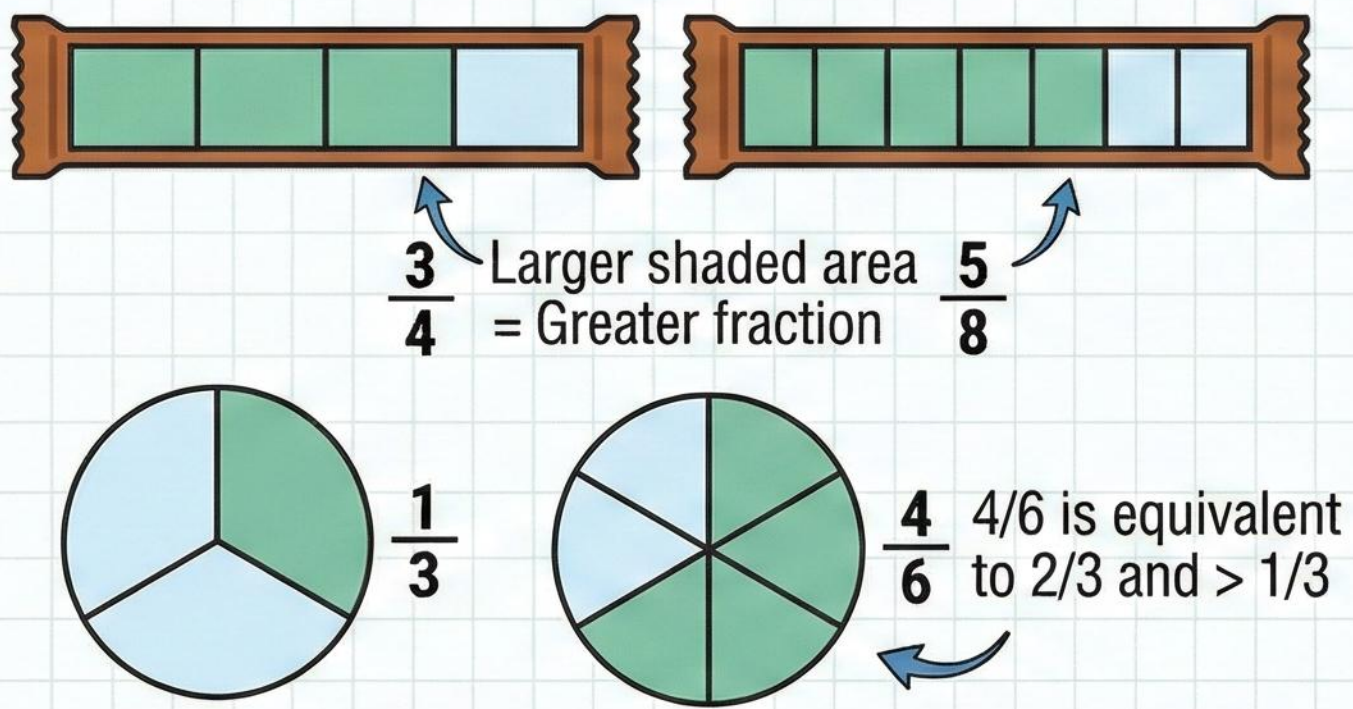
NURTURE YOUR SKILLS! Practice Makes Progress.



COMPARING FRACTIONS: STRATEGIES & VISUALS



1. VISUAL MODELS: SEEING THE SIZE



2. SAME DENOMINATORS: COMPARE THE TOPS

$$\frac{5}{8} \longrightarrow \frac{3}{8}$$

Compare numerators: $5 > 3$

So, $\frac{5}{8} > \frac{3}{8}$

CROSS MULTIPLICATION:
For a/b and c/d , if $a*d > b*c$, then $a/b > c/d$.

NUMBER LINES:
Place fractions on a number line; the one further right is larger.

SAME NUMERATOR:
Smaller denominator = Larger fraction.

EQUIVALENT FRACTIONS:
Look different, same value (e.g., $1/2 = 2/4$).

QUICK TIPS & TRICKS

3. DIFFERENT DENOMINATORS: FIND COMMON GROUND

$$\frac{2}{3} + \frac{3}{4} = >$$

A Find Least Common Denominator (LCD)
LCD for 3 and 4 is 12

B Convert to Equivalent Fractions
 $\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$
 $\frac{3}{4} = \frac{3 \times 3}{4 \times 3} = \frac{9}{12}$

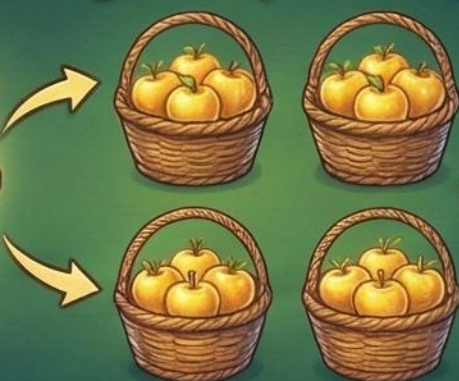
C Compare Numerators
 $\frac{8}{12} 8 < 9 \frac{9}{12}$
Therefore, $\frac{2}{3} < \frac{3}{4}$



THE ORCHARD OF DIVISION: SHARING & GROUPING

1

THE SEED: What is Division? (Splitting into Equal Groups)



$$\rightarrow 12 \div 4 = 3$$

Dividend: 12

Divisor: 4 Quotient: 3

Sharing a total into equal parts.

2

THE ROOTS: Visual Models

(a) Sharing (Baskets & Friends)



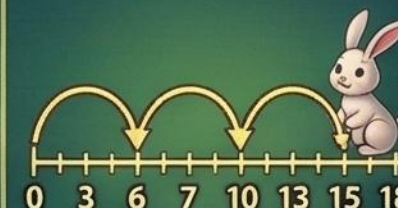
20 shared by 4 = 5 each
Equal distribution.

(b) Arrays (Rows & Columns)



24 $\div$ 4 = 6 rows
24 $\div$ 6 = 4 columns
Organized grid.

(c) Number Line (Jumps Back on a Path)



18 $\div$ 6 = 3 jumps
Equal steps backward.

3

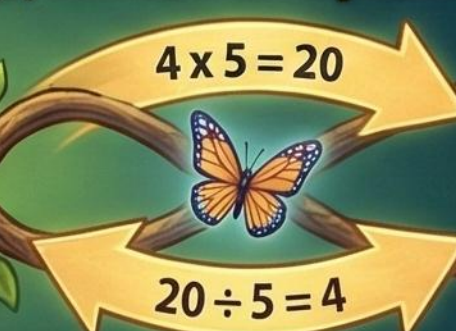
THE BRANCHES: Key Terms & Properties (Rules of Sharing)

(a) The Parts of Division



Dividend $\div$ Divisor = Quotient

(b) Inverse of Multiplication



Doing and undoing.
(e.g., If $4 \times 5 = 20$, then $20 \div 5 = 4$)

4

THE FRUITS: Real-World Harvest (Applications)

Sharing Food



8 slices $\div$ 2 people
= 4 slices each

Calculating Rate



Speed = 120 miles
 $\div$ 2 hours = 60 mph

Unit Price



Cost per box
= $\$3 \div 6 = \0.50

5

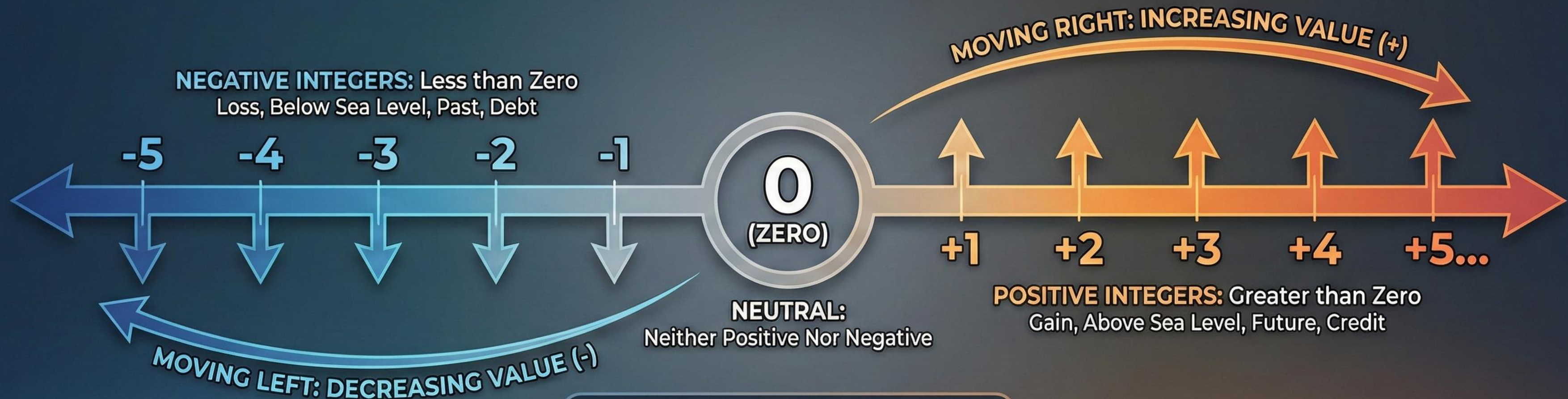
THE TOOL: The Harvester's Chart (Division Table)

1	1	3	4	5	6	7	8	9	100
10	12	13	14	15	16	17	18	19	20
30	21	22	23	25	26	27	28	29	30
40	41	42	43	45	46	47	48	49	50
70	51	52	53	55	56	57	58	59	60
80	71	72	73	75	76	77	78	79	80
100	10	32	30	55	60	70	80	99	10

A helpful reference for quick quotients.

HARVEST YOUR UNDERSTANDING!
Practice Makes Perfect.

THE WORLD OF INTEGERS: WHOLE NUMBERS, POSITIVE & NEGATIVE



REAL-WORLD EXAMPLES (NEGATIVE)



-10°C

Temperature, Depth, Owing Money



-200m



-\$50

OPERATIONS & PROPERTIES

OPPOSITES



Additive Inverses: Sum is 0

ABSOLUTE VALUE

$$|-4| = 4 \quad \text{Distance from Zero (Always Non-Negative)}$$



$$|+4| = 4$$

Distance from Zero (Always Non-Negative)

REAL-WORLD EXAMPLES (POSITIVE)



+25°C



+1500m



+\$100

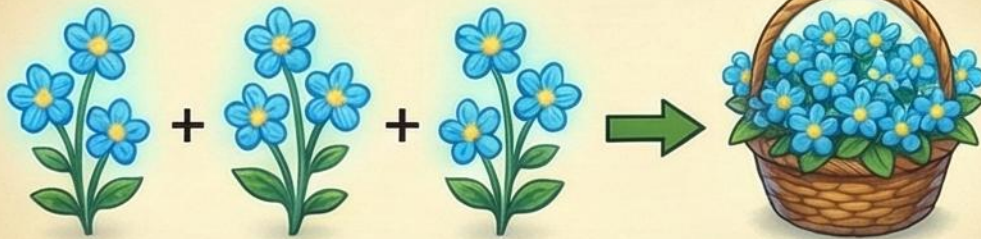
Temperature, Elevation, Saving Money

INTEGERS = {..., -3, -2, -1, 0, 1, 2, 3, ...} | THE BUILDING BLOCKS OF ARITHMETIC & BEYOND

THE GARDEN OF MULTIPLICATION: GROWING WITH NUMBERS

1

THE SEED: What is Multiplication? (Repeated Addition)



$$3 + 3 + 3 + 3$$

$$4 \times 3 = 12$$

Adding the same number multiple times.

2

THE ROOTS: Visual Models

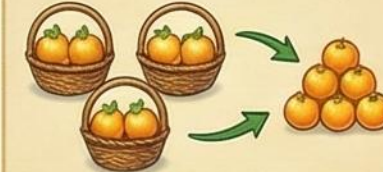
(a) Arrays (Rows & Columns)



$$5 \times 4 = 20$$

Organized grid.

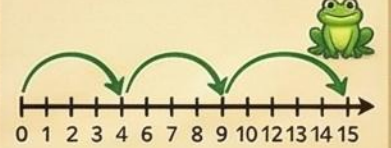
(b) Equal Groups (Bundles & Baskets)



$$3 \text{ groups of } 2 = 3 \times 2 = 6$$

Same number in each set.

(c) Number Line (Jumps on a Path)



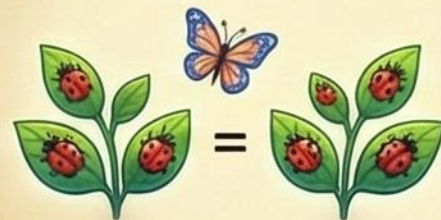
$$3 \text{ jumps of } 5 = 3 \times 5 = 15$$

Equal steps forward.

3

THE BRANCHES: Key Properties (Rules of Growth)

(a) Commutative Property (Order Doesn't Matter)



$$a \times b = b \times a$$

(e.g., $2 \times 3 = 3 \times 2$)

(b) Associative Property (Grouping Can Change)



$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$
$$(a \times b) \times c = a \times (b \times c)$$

(e.g., $(2 \times 3) \times 4 = 2 \times (3 \times 4)$)

(c) Distributive Property (Breaking it Down)



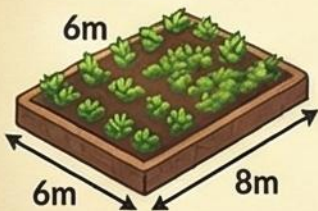
$$4 \times 12 = 4 \times 2$$
$$a \times (b + c) = (a \times b) + (a \times c)$$

(e.g., $4 \times 12 = 4 \times 10 + 4 \times 2$)

4

THE FRUITS: Real-World Harvest (Applications)

Calculating Area



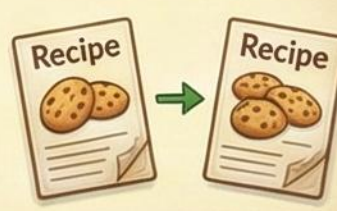
$$\text{Area} = 6 \times 8 = 48 \text{ sq m}$$

Total Cost



$$\text{Cost} = 5 \times \$2 = \$10$$

Scaling Recipes



Double the batch
= multiply by 2

5

THE TOOL: The Gardener's Chart (Multiplication Table)

	1	2	3	5	6	7	8	9	10
1	1	2	3	5	6	7	8	9	10
2	2	4	6	10	12	14	16	18	20
3	3	6	9	15	18	21	24	27	30
4	4	8	12	20	24	28	32	36	40
5	5	10	15	25	30	35	40	45	50
6	6	12	18	30	36	42	48	54	60
7	7	14	21	35	42	49	56	63	70
8	8	16	24	40	48	56	64	72	80
9	9	18	27	45	54	63	72	81	90
10	10	20	30	50	60	70	80	90	100

A helpful reference for quick products.

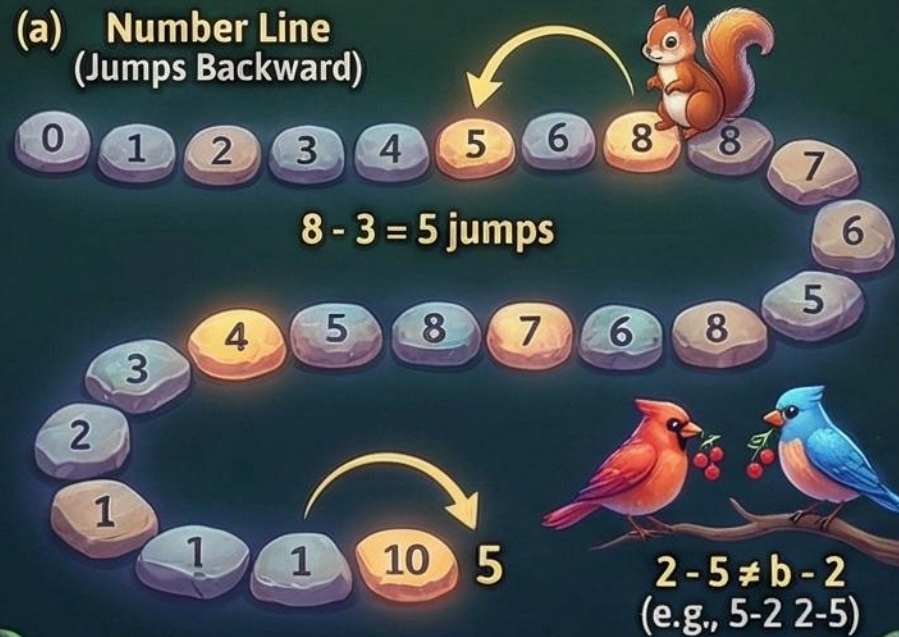
CULTIVATE YOUR KNOWLEDGE!
Practice Makes Progress.

THE TREE OF SUBTRACTION: HARVESTING THE DIFFERENCE

1

THE BRANCHES: Mogings Backward

(a) Number Line
(Jumps Backward)



1

THE ROOTS: What is Subtraction? (Taking Away)



3

THE BRANCHES: Key Properties (Rules of Harvest)

(a) Not Commutative
(Order Matters)

(b) Inverse of Addition
(Undoing)



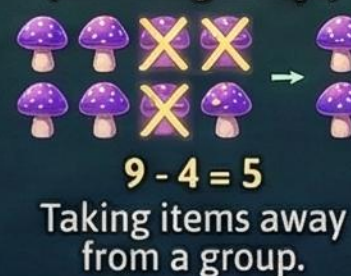
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THE TRUNK: Visual Models

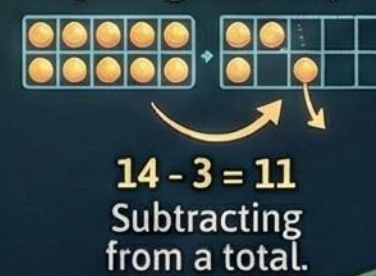
(a) Number Line
(Jumps Backward)



(b) Counters
(Removing Groups)



(c) Ten-Frames
(Taking from Ten)



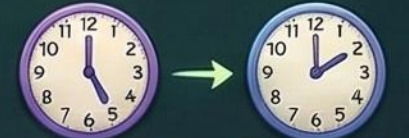
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THE FRUITS: Real-World Harvest (Applications)

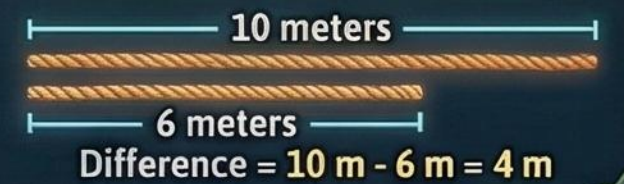
Spending Money



Time Elapsed



Difference in Length



5

THE TOOL: The Harvester's Chart (Subtraction Table)

A helpful reference for quick differences.

	1	10	4	6
4	5	9	7	5
6	7	6	8	6
8	4	5	6	6
10	4	2		
12	1	3	6	9
16	5	4	7	10

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Practice Makes Progress.